

Ophthalmic Optics

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INTRODUCTION

Ophthalmic optics deals with optical elements that are employed to improve vision. The necessity for such improvement may stem from 1) the presence of a refractive error or any other visual defect; 2) a particular or professional demand for increased retinal image size; 3) increased image contrast. Protective filters are also traditionally included within the scope of ophthalmic optics. There are a large number of optical components that can be used to improve vision: spectacle lenses and filters, contact lenses, intraocular lenses, and instruments for low vision. They can be refractive, diffractive, or both. This work concentrates mainly on the properties, design, manufacture, and adaptation of spectacle lenses, which account for the largest number of users.

The foundations for the design and manufacture of modern single vision, bifocal, and trifocal lenses were laid out at the end of the 19th century. Nowadays, intensive research is conducted in the design and manufacture of progressive lenses, design of contact lenses, design of contact and intraocular multifocal lenses, and plastic materials for spectacle and other types of lenses.

OVERVIEW

Materials for ophthalmic optics are mainly glass and plastics. Contact and intraocular lenses are all made of plastic. For spectacle lenses, glass is also being surpassed by plastics, which are usually coated to improve their scratch resistance. In the last few years, ophthalmic lenses have been increasingly considered as compound systems in which surface geometry, material, and a more or less complex coating structure are joined together to give the overall lens performance.

In this work, we begin with the description of the eye–lens system. This analysis is necessary to fully understand the specific properties and constraints of the ophthalmic lens design. This design is presented as the two main formats of ophthalmic lenses. Aspheric surfaces are also taken into account. A proper analysis of the fabrication of ophthalmic lenses is likewise included. Lenses for presbyopia are presented in their typical formats—from monofocals to progressive lenses. Finally, the “Conclu-

sions” section provides some hints about the importance of these simple and useful optical systems.

THE EYE–LENS SYSTEM

Ophthalmic optics does not mean anything without taking into account the characteristics of the human eye. The study of the design and performance of the ophthalmic optical systems requires the knowledge of the optics of the eye and utilization of its anatomical and physiological characteristics and limitations.

The Eye as an Optical System

The eye can be considered as an optical system with a positive power of about 58 D.^[1–3] It has two main refractive elements, the cornea and the lens. The cornea bulges in the front of the eyeball, and because its first surface is in contact with air, it bears most of the power of the eye (about 45 D). The eyeball has a mean length of 24 mm, and the image is formed at the interior of the back side, where the retina is found. The aqueous humor, which has a refractive index of 1.336, is located in the midst of the cornea and the lens. The volume behind the lens is filled with vitreous humor, with index 1.337. The power of the lens is not fixed, and it can expand surface curvature and power via the ciliar muscles surrounding it. This process is called accommodation, and serves to bring near objects into focus. The pupil of the eye is located 3 mm behind the front vertex of the cornea. Its diameter ranges from 2 to 8 mm, and it is influenced by lighting conditions, age, drugs, and health status.

Visual acuity^[4] is defined as the inverse of the smallest angle subtended by two points so they can be resolved. Visual acuity is not constant along the field imaged by the optics of the eye. The highest density of photoreceptors, and therefore the highest visual acuity, is located in a very narrow region known as the fovea. We distinguish colors, read (among other highly specialized tasks), and scrutinize the world by using this small part of the retina. Therefore, when we say the image is focused onto the retina, we really mean that it falls on the fovea, where it can be better detected, recognized, and identified by the visual cortex in the brain. The outer retina develops other



important missions in our spatial perception of the surrounding visual environment.^[2] In order to extend the small angle subtended by the fovea to a wide field with high resolving power, the eye rotates in different directions within different ranges. The location of the center of rotation relates with the anatomic structure of the eyeball and the surrounding muscles.^[5] Although each movement has its own center of rotation, it is possible to define a mean center of rotation, located, on average, 15 mm behind the corneal apex.

Ammetropies and Refractive Error

Emmetropia is defined as the condition for which the relaxed eye (without accommodation) images a distant object onto the retina (fovea).^[6] We say that the eye suffers from refractive error when it fails to bring into focus the image of a distance object. The condition in which a refractive error occurs is called ammetropia.^[1,2,6] Let us define the remote point as the conjugate point of the fovea in a relaxed eye. The refracting error, R , is then defined as the inverse of the distance from the principal plane of the eye to the remote point. The conjugate point of the fovea moves when the eye accommodates. It reaches the so-called near point when the eye attains its maximum accommodation. The range between the remote and near points is called the range of clear vision. All points within it are accessible via accommodation. Amplitude of accommodation is defined as the difference between the refractive error and the inverse of the distance between the principal point of the eye and the near point.^[7]

Ammetropia can be classified into four different types: two forms of spherical ammetropia (myopia and hyperopia), astigmatism, and presbyopia.^[6,8]

Myopia: The image of a distant object forms before the retina because the eye is too powerful, too large, or both.

Refractive error is negative and the remote point is located in front of the eye. Near point is also located in front of the eye, at a smaller distance from it.

Hyperopia: The image of the distant object forms behind the retina as a result of any of the following reasons: 1) the eye does not attain enough power, 2) the eye is too small, or 3) combination of both. In case the refractive error is smaller than the amplitude of accommodation, the hyperopic eye may bring the images of distance objects into focus via accommodation. The remote point of the hyperopic eye is located behind (to the right of) the same. The near point can be located at the right of the remote, when amplitude of accommodation is smaller than the refractive error. Otherwise, it is located at a finite distance in front of the eye.

Astigmatism: The eye shows different powers at different meridian planes. We will assume that the meridian planes with maximum and minimum power are orthogonal (regular astigmatism). These are called the principal meridians. The power difference between them is termed the astigmatism of the eye. When characterizing astigmatism, aside from the value of the difference between powers, it is also necessary to provide the orientation of any of the principal meridians.

Presbyopia: The amplitude of accommodation reduces with age and along with it, the ability to focus near objects. When amplitude of accommodation reduces below 3 D (which usually happens between 40 and 50 years of age in Europe), comfortable reading at 33 cm is no longer possible, and ophthalmic compensation is necessary for activities using near vision.

When the eye rotates, the remote point, locked to the visual axis, also rotates. Assuming a well-defined rotation center, the locus of points that correspond with remote points for different sight directions is a sphere called remote point sphere. The center of this sphere coincides

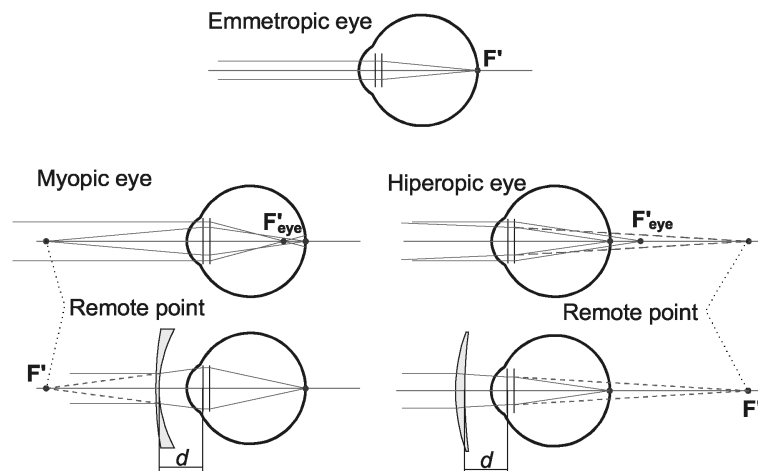


Fig. 1 Principle of compensation and lens adaptation.

with the rotation center of the eye. As the eye accommodates, the radius of the remote point sphere changes accordingly with eye ametropia. For myopic eyes, the radius decreases as accommodation increases. The same occurs for hyperopic eyes with accommodation values greater than the refractive error. For hyperopic eyes with accommodation values smaller than the refractive error, the radius of the remote point sphere increases with accommodation. In any case, the accommodation always causes the remote point sphere a movement to the right (light travels from left to right). When accommodation reach its maximum, the sphere of conjugate points turns into the near point sphere.

In astigmatic eyes, there are two remote and two near points, one for each of the principal meridians. Accordingly, there are two remote point spheres and two near point spheres. Fig. 1 shows the situation of the remote point and the near point for a myopic and a hyperopic eyes.

The Principle of Compensation

By using those previously defined concepts, we can state the principle of compensation as follows:^[6,8] a lens placed in front of the eye compensates for its ametropia when the image focal point of the lens coincides with the remote point of the eye.

This compensation principle applies only for the situation in which

- a) the lens axis passes through the rotation center of the eye, and
- b) the lens axis coincides with the eye visual axis.

The lens should be fitted in such a way that the first condition, which does not depend on the eye angular position, is satisfied. That way, astigmatism resulting from central oblique incidence will be avoided. The position in which a person aims at infinity, with visual axis parallel one to the other and to the floor, is defined as primary sight position.

At oblique sight directions from the primary position, the second condition is not satisfied, and a deeper (nonparaxial) study will be necessary to understand lens performance. The result of this analysis will determine the optimum lens design, which guaranties an oblique behavior as similar as possible as the paraxial one. In the rest of this section we will present the characteristics of the lens–eye coupling, which can be adequately described within the paraxial approximation.

Compensation of Spherical Ametropia

Let us define the back vertex power of the lens as the inverse of the distance from the back surface vertex to the

image focal point. In terms of surface refracting powers, F_1 and F_2 , the back vertex power is given by^[8–10]

$$F_V = \frac{F_1 + F_2 - (t/n)F_1F_2}{1 - (t/n)F_1}$$

where t is the central thickness and n is the refraction index of the material. Let us assume that the distance from the back surface vertex of the lens to the object principal plane of the eye is d . According to the compensation principle, the back vertex power required to compensate a refractive error R is given by^[6,8,9]

$$F_V = \frac{R}{1 - dR}$$

For refractive errors smaller than 4.5 D, the term dR becomes negligible and refractive error and back vertex power of the correcting lens are practically the same. For larger values of the refractive error, back vertex power differs from it by more than 0.25 D, and the difference must be taken into account.

Compensation of Astigmatism: The Dioptric Power Matrix

For an ophthalmic lens used to compensate astigmatism, the compensation principle should be applied in the two orthogonal planes where the maximum and minimum powers of the eye appear.^[8,9,11] Let us consider R_α and R_β the principal values of the refractive error, at meridians α and $\beta = \alpha + 90^\circ$. The principal powers of the astigmatic compensating lens will be $F_{V\alpha} = R_\alpha / (1 - dR_\alpha)$ and $F_{V\beta} = R_\beta / (1 - dR_\beta)$. It is customary to name any of these back vertex powers as sphere (S), and the difference between them as the cylinder (C). For example, we can select $S = F_{V\alpha}$, so that $C = F_{V\beta} - F_{V\alpha}$. Of course, we can also set $S = F_{V\beta}$ and $C = F_{V\alpha} - F_{V\beta}$. With these definitions, astigmatic lenses can be understood as the thin superposition of two elements—one purely spherical, with power S , and the other purely cylindrical, with power C —and whose axis is oriented along the meridian with power equal to the sphere. Complete description of astigmatic power requires three parameters. They could be the orientation of one of the principal meridians and their powers. Instead, it is more common to use the sphere, cylinder, and cylinder axis orientation. These three components are usually written as $[S, C \times \alpha]$.

Although the use of sphere and cylinder is widely employed by ophthalmologists, optometrists, and practitioners, in a technical depiction of ophthalmic lenses, it is far more powerful to employ a matrix description of astigmatic power.^[11–17] Whereas sphere and cylinder still refer to power along principal meridians, power compo-



nents referred to fixed coordinate axis can be grouped into a matrix form (tensor), so that any mathematical transformation or manipulation is easily written and performed. The dioptric power matrix, $\mathbf{F}$, is defined as

$$\mathbf{F} = \begin{pmatrix} E + C \sin^2 \alpha & -C \sin \alpha \cos \alpha \\ -C \sin \alpha \cos \alpha & E + C \cos^2 \alpha \end{pmatrix}$$

and it accounts for all the thin lens properties of any astigmatic lens. The dioptric power matrix is just the 2×2 submatrix at the bottom left of the 4×4 ABCD matrix corresponding to the thin astigmatic lens.

For example, if we consider two lenses with power matrix $\mathbf{F}_1$ and $\mathbf{F}_2$, their superposition is corresponded with a matrix $\mathbf{F}_1 + \mathbf{F}_2$. The local thickness of the lens, $t(x,y)$, is given by

$$t(x,y) = t_0 - \frac{(x \ y) \mathbf{F} \begin{pmatrix} x \\ y \end{pmatrix}}{2(n-1)}$$

where t_0 is the central thickness and n is the refraction index of the lens material.^[18–22] Matrix formulation of dioptric power also allows for rigorous statistical treatment of clinical data, as extensively demonstrated by W. Harris and others.^[23–30]

Lens Adaptation

We have already shown the relationship between lens power and refractive error. Now, we are going to describe the way a lens couples with the eye, modifying the way the eye perceives images. All the effects derived from this optical coupling are grouped under the concept of *lens adaptation*. The monocular effects are: prismatic effect, modification of the field of view, lens magnification, and ophthalmic lens aberrations. The first three effects can be analyzed in the paraxial regime.

Prismatic effect

From all the rays refracting in the lens and entering into the pupil of the eye, as shown in Fig. 2, only a narrow ray bundle contributes to the foveal vision. This bundle is mainly characterized by the chief ray, which passes through the rotation center of the eye, and the pupil of the eye, which determines the bundle diameter. The compensating lens does not only change the bundle vergence in order to bring it to a focus at the remote point. It also changes the direction of the chief ray, and henceforth, of the whole ray bundle. As a result of this change in direction, the eye has to rotate at an angle from the position in which it could observe the object without the compensating lens. The angle rotated by the eye (from the situation without the lens to the situation with the lens) is

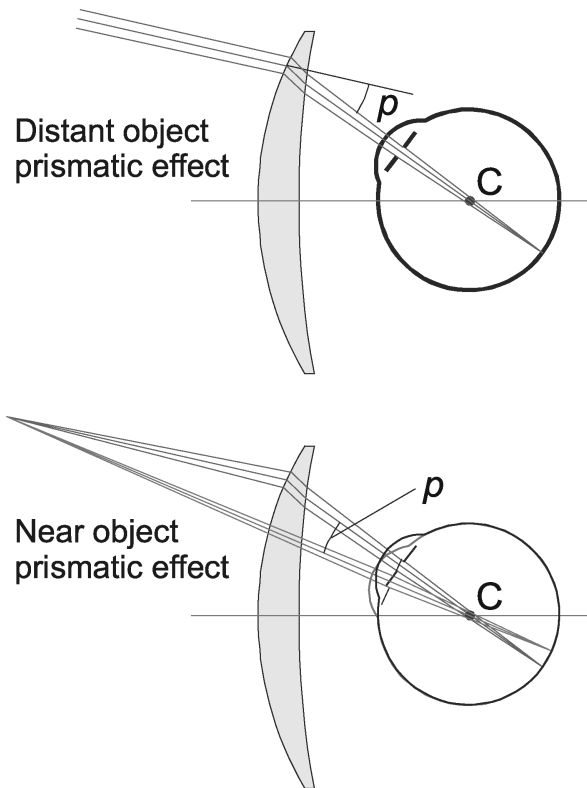


Fig. 2 Prismatic effect for a distant and near object.

called the prismatic effect. Sight direction is characterized by two direction cosines. The prismatic effect will also have two components that will equal the differences between direction cosines before and after the compensating lens is introduced. The two components of the prismatic effect are arranged in a vector

$$\mathbf{p} = \begin{pmatrix} p_x \\ p_y \end{pmatrix}$$

The prismatic effect introduced by a lens with matrix power $\mathbf{F}$ at a point of coordinates $r = (x,y)$ is given by

$$\mathbf{p} = -\mathbf{F}\mathbf{r}$$

which is called the matrix Prentice's law.^[12,31,32] Prismatic effect at a point (x,y) becomes larger as the distance from the point to the optical center increases. Furthermore, prismatic effect is proportional to lens power. The prismatic effect introduced by a spherical lens has radial geometry. The prismatic effect introduced by a purely cylindrical lens is perpendicular to the cylinder axis.

In some instances, prismatic effect is prescribed in order to compensate for a convergence or binocular visual problem. The required prismatic effect may sometimes be

generated by lens decentring. The decentring δr , required to produce a determined prismatic effect p , is given by

$$\delta r = -F^{-1}p$$

Modification of the field of view

The dynamic field of view (the scope accessible to foveal vision via eye rotation) is modified by the presence of the frame and the ophthalmic lens (Fig. 3).^[8,33] The apparent field of view is defined in terms of the frame size, and is given by

$$\varphi_a = \arctan(hL'_2)$$

where h is the distance from the frame border to the visual axis in primary position, and L'_2 is the inverse of the distance from the back surface vertex to the rotation center of the eye, l'_2 . The real field of view takes into account the modification of the apparent field of view caused by the lens prismatic effect. The user experiences this visual field, and is given by

$$\varphi_r = \arctan[h(L'_2 - F_V)]$$

Hyperopsics experience a decrease of the field of view, whereas myopsics experience an increase of the same.

Lens magnification

Ophthalmic lens also modifies the size of the retinal image if compared with that of the uncorrected eye.

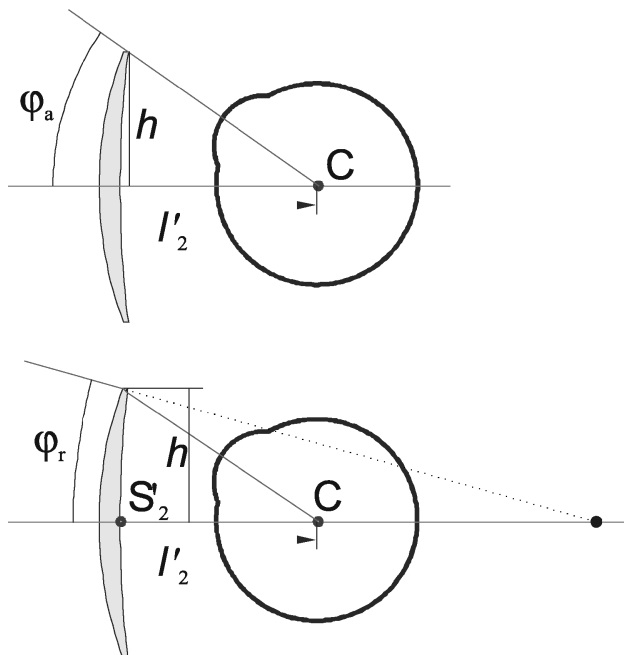


Fig. 3 Field of view through an ophthalmic lens.

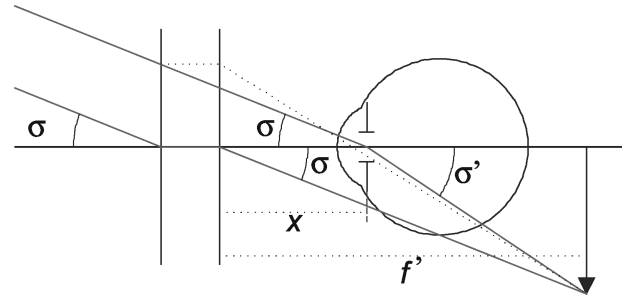


Fig. 4 Spectacle magnification.

Spectacle magnification (SM) is defined as the quotient between the size of the retinal image in the compensated eye and the size of the (blurred) retinal image in the uncompensated eye.^[8,33–35] SM is easily computed through the expression,

$$SM = \frac{1}{1 - xF}$$

where x is the distance from the image principal plane of the lens and the front pupil of the eye, and F is the lens power (Fig. 4). If we write this equation in terms of more handy parameters (back vertex power and distance from the back surface vertex to the corneal vertex, d) we obtain

$$SM = \frac{1}{1 - (e/n)F_1} \frac{1}{1 - dF_V}$$

So, two terms contribute multiplicatively to spectacle magnification. The first term, called the shape factor, does not depend on lens power. The second term, called the power factor, neither depends on lens form factor nor on thickness. The shape factor may be used to equalize spectacle magnification in both eyes when aniseikonia (different image sizes in each eye) occurs. Lenses that use this principle are called isekonic lenses.^[35]

Ophthalmic lens aberrations

We previously stated the conditions for which the paraxial description of the lens–eye system is adequate. At oblique sight directions, the lens optical axis does not coincide with the visual axis, and aberrations have to be taken into account.^[8,36] The typical situation is depicted in Fig. 5. The rays that impinge onto the fovea after refraction through the ophthalmic lens form a narrow pencil whose transversal extension is limited by the pupil of the eye. The ray passing through the center of the pupil and the rotation center of the eye is called the chief ray. It coincides with the axis of the ray bundle under consideration.

At oblique sight directions, the principle of compensation must be slightly modified. In order to adequately



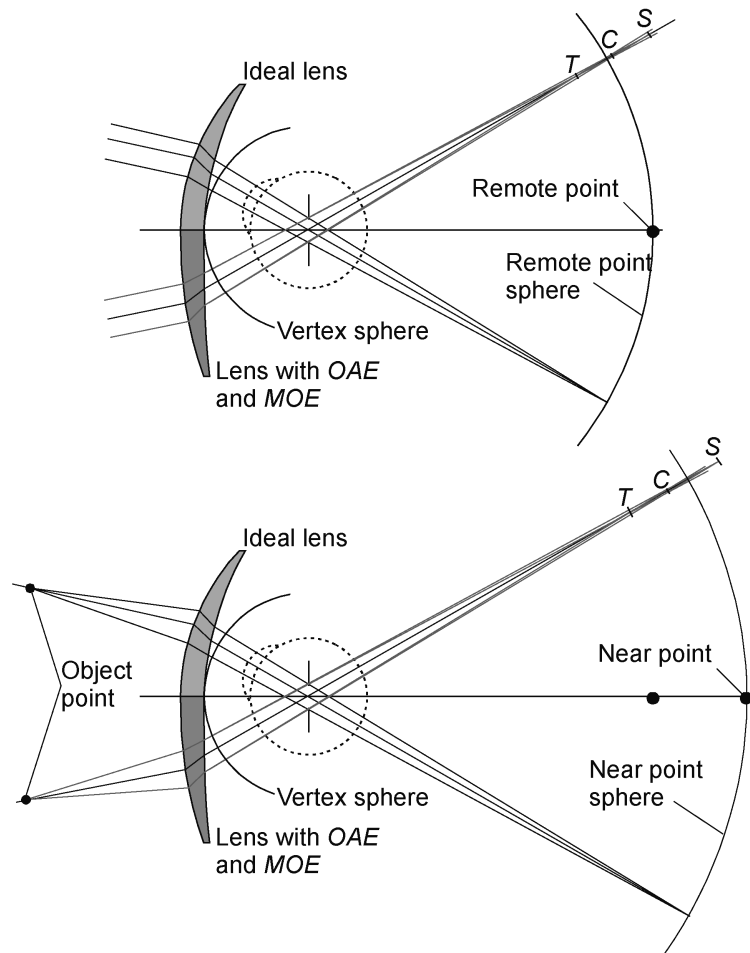


Fig. 5 Oblique astigmatism error (OAE) and mean oblique error (MOE) for a positive lens in the cases of distant and near objects.

compensate for the refractive error at an oblique sight direction, the ray bundle around the chief ray that refracts through the lens and goes through the eye pupil should focus onto the remote point sphere, exactly at the remote point of the eye for that oblique direction.

In this situation, only those aberrations relating to thin pencil of rays will be noticeable. Because of the small aperture of the ray bundle, spherical aberration and coma are not taken into account (they will be noticeable only for very high refractive errors). There will be distortion, although this aberration does not modify visual acuity but image position. Distortion is apparent when the whole field of view is taken into account. In any case, the brain adapts easily to distortion so the user ignores it after the first few days of wearing spectacles. Oblique astigmatism, which generates two focal lines, and field curvature, which generates the Petzval surface, affect visual acuity and are the aberrations to be corrected. Because of the particular characteristics of the lens–eye system, these two classical aberrations have to be redefined.

First, powers and focal lengths will be defined along the chief ray, and with reference to the vertex sphere. This is the sphere with center at the rotation center of the eye and tangential with the ophthalmic lens back surface. Let us take A as the point where the chief ray intersects the vertex sphere, and T and S as the tangential and sagittal focal lines of the refracted pencil, respectively. Finally, we take C as the point at which the disk of least confusion of the astigmatic pencil is located. The tangential, sagittal and mean oblique powers, F_T , F_S , and F_C are defined respectively as

$$F_T = \frac{1}{AT}, \quad F_S = \frac{1}{AS}, \quad F_C = \frac{1}{AC}$$

and it can be demonstrated that $F_C = (F_T + F_S)/2$.

The oblique astigmatic error (OAE) is defined as the difference between the oblique tangential and sagittal powers. The mean oblique error (MOE) is defined as the difference between the back surface power and the mean oblique power. As can be observed, the ophthalmic lens

should present the field curvature so that the compensation principle at oblique sight directions is satisfied. The MOE then stands for the difference between the actual field curvature and the rate required for the refractive error that the lens compensates for. An ideal ophthalmic lens should have negligible values of oblique errors for all values of the obliquity angle, u'_2 .

Chromatic aberration will always be present in ophthalmic lenses. Transverse chromatic aberration (TCA), in particular, has the largest impact on visual acuity at oblique sight directions. Unfortunately, there is no way to control it except for the selection of materials with the highest Abbe number.

OPHTHALMIC LENS DESIGN

This section will show the relationship between the shape of an ophthalmic lens and the oblique errors of the same, and how they can be minimized.^[8,36,37] The parameters that completely describe a thick lens with spherical surfaces include the two radius of curvature of the surfaces, the index of refraction, the thickness, and the transversal diameter. The election of the refraction index is limited to the use of a few selected and widely available plastic and glass materials. The central thickness is also limited by constructive constraints. For negative powers, the central thickness needs to be large enough to guarantee security. For a positive lens, the central thickness is determined by the transverse diameter and the expected value of the edge thickness. In any case, the variation range is within a few millimeters. Lenses are available in standardized diameters (typically 50–80 mm), which allow the users to fit the lenses in a large variety of frames properly. We conclude that these parameters cannot be left free in order to reduce aberration. The relation between the required back vertex power and the lens parameters establishes a constraint between the two surface radii so that only one free parameter is left—which can be any of the surface radii or, more commonly, any of the two surface powers of the lens. Without losing generality, let us assume that the back surface power is left free in order to optimize lens performance. The objective of the lens designer will be to find the value of the back surface power that best fit the designer's criteria for oblique errors correction (see next section for an explanation of these criteria).

In order to compute for the oblique powers, it is necessary to determine the angles and points of incidence of the chief rays on the two lens surfaces first. To do that, a backward exact ray tracing of the chief ray is carried out. Once the trajectory of the chief ray is completely known, Coddington equations can be applied to each surface so that the position of the tangential and sagittal focal lines

are determined. Finally, some geometrical computation to find the intersection of the chief ray with the vertex sphere leads directly to the oblique powers and henceforth to the oblique errors. Full analysis can be programmed in a few lines of code so that extensive computation of oblique errors as a function of the back (or front) surface power can be easily carried out.

Best Form Lenses. Point-Focal and Percival Solutions

It should be recalled that although oblique astigmatism is zero for the primary position, it increases with the angle of incidence. For a given sight direction, there will be a determined amount of astigmatism which will depend on the lens shape factor. If absolute value of the lens power is not too large, a shape factor can be found for which the astigmatism generated at the first surface is cancelled out exactly by the astigmatism at the second refraction. A lens with this property is called point-focal (with punctual focus). This condition can only be met for a determined sight direction. Fortunately, astigmatism remains negligible for smaller angles, so it can be assumed that in a point-focal lens the tangential and sagittal shells merge onto the unique Petzval surface, which is generated by the field curvature. Usually, the radius of curvature of the Petzval surface of a point-focal lens does not coincide with the radius of the far point sphere. Henceforth, the focus will not fall onto the far point sphere for an oblique sight direction, and the lens will suffer from power error. This error is negative for myopic people and positive for hyperopic people, leading in all cases to under-correction at oblique incidence. Hyperopic persons will be able to cancel this error via a small amount of accommodation (usually less than half a diopter).

Instead of looking for lenses with punctual focus, one could try to eliminate the power error—that is, to bring the disk of least confusion onto the far point sphere, at least for one oblique sight direction. This was proposed by English ophthalmologist A. Percival (1914), who noticed that the blur patch on the retina was smaller when a lens was made to satisfy his criterion than when it was made to be point-focal. For example, consider a +5.00 D lens, made with crown glass ($n = 1.523$), and assume that it will be made with a central thickness of 3.5 mm, and that it will be worn so that $l'_2 = 27$ mm. Exact ray tracing for a 35° rotation of the eye shows that with a back refracting power of -6.50 D, the lens will be point-focal, whereas a back refracting power of -4.00 D will enable the lens to satisfy the Percival criterion. If the diameter of the pupil is assumed to be 4 mm, the size of the blur patch for the point-focal design will be 0.0255 mm (13 foveal cones). For the Percival design, the blur patch coincides with the



disk of least confusion, which for this lens has a diameter of 0.0135 mm (seven foveal cones). This result led Dr. Percival to propose the power error free lens form (or bending, as it is usually named in ophthalmic optics) as the optimum design. Indeed, the Percival form is not as advantageous as could be derived from the previous result. First, the disk of least confusion is not an evenly illuminated blur patch (as it is in the case with point-focal lenses), and the distribution of light on it is more complicated. Second, hyperopic may bring to focus the blur patch of a point-focal lens via a small amount of accommodation. Third, the Percival condition is more critical than the point-focal condition to variations of the design parameters (fitting distance and object vergence). Finally, variations in accommodation cause the blur patch to lose its rotational symmetry in Percival lenses, whereas circularity of the blur patch is preserved in point-focal lenses. For all these reasons, point-focal forms were mainly chosen for the manufacture of ophthalmic lenses for most of the 20th century.

A design compromise that is widely used nowadays consists of making lenses free from tangential error^[8]—that is, lenses for which the tangential shell coincides with the far point sphere. The astigmatism of such a lens is quite small (about half of the astigmatism of a Percival form) and the design is stable relative to the variation in fitting distance and object vergence. In relation to the fitting distance, it is assumed that variations of up to ± 6 mm, from the mean value of 27 mm, may arise. When the fitting distance increases, the lens tends to become point-focal, and when the fitting distance decreases, the lens tends to take a Percival form. In relation to object distance, a tangential error-free form designed for distance vision behaves quite well for near vision. In changing from distance to near vision, power error remains remarkably similar, and astigmatism simply changes its sign.

In all these cases, aberration elimination (regardless if it is OAE, mean oblique power error, or oblique tangential error) is accomplished via meniscus-shaped lenses. For these lenses, any of these aberrations at each surface presents opposite sign, so the proper lens bending may bring the aberration absolute values to be equal on each surface and are henceforth compensated. Nevertheless, compensation of aberrations with spherical surfaces presents some limitations. On the one hand, this compensation effect only works in a determined range of lens power, which depends on the aberration to be compensated, the fitting distance, and the refraction index of the lens material (central thickness has very little effect on aberration). The powers for which aberration can be compensated usually range from high-powered negative lenses (about -25 D) to medium-powered positive lenses (about 7 D). It therefore follows that it is not possible

to have high-powered spherical lenses free from oblique errors—like, for example, the lenses for correcting aphakia. On the other hand, given a determined power, spherical surfaces yield two different bendings for which any of the oblique errors may be compensated. The steeper bending is called the Wollaston form, and it uses very high refractive powers (typically with back surface powers of about -20 to -35 D). Wollaston lenses are so curved that they are difficult to manufacture, and difficult to cut in order to fit them to a frame. Indeed, they are also quite unsightly. In his experiments, Wollaston discovered that these lens forms were free from astigmatic error, and he called them periscopic. The shallower bending is called the Ostwalt form. They have more distortion than those with Wollaston bending, but they behave similarly with respect to oblique errors. Although modern spherical lenses are made with bending as close as possible to the Ostwalt form, the prevalent aesthetic criteria nowadays demands for even flatter lens designs.

The third-order approach to the calculation and cancellation of the oblique astigmatism, or the power error,^[37] has provided analytical solutions for the point-focal and Percival formats in the past. For a thin lens, when these two aberrations are expressed in terms of the total power of the lens, F' , and one of the surface powers of the lens, the loci of the formats having zero oblique astigmatism (or zero error power) are on two ellipses called Tscherning ellipses. Once the surface power of one of the surfaces is known, the other is easily found by using the thin lens assumption.

The location and form of the Tscherning ellipses depend on the index of refraction, and the distance from the back vertex of the lens to the center of rotation. These ellipses have been plotted in Fig. 6, where the map of the aberrations is analyzed. An initial view of this graphic shows that, for a given value of total power (i.e., for a given refractive compensation), there are two possible formats that cancel the aberration under analysis. One of the formats has lower absolute value of the surface powers than the other format. This means that the surfaces will be flatter in one of the formats. This flatter solution is the Ostwald format, and the other is the Wollaston solution. Both of them should be equivalent within this third order and thin lens condition.

An exact calculation based on real ray-tracing can be used to know the formats having a given value of aberration. These formats depend now on the actual parameters of the lens: index of refraction, central thickness, transversal diameter, and radii of curvature of the surfaces. They also depend on the consideration of far vision or near vision. In Fig. 6, we show a contour plot of the oblique astigmatism aberrations for far vision. The contour lines correspond with values of 0.125, 0.25, and 0.5 D of aberration.

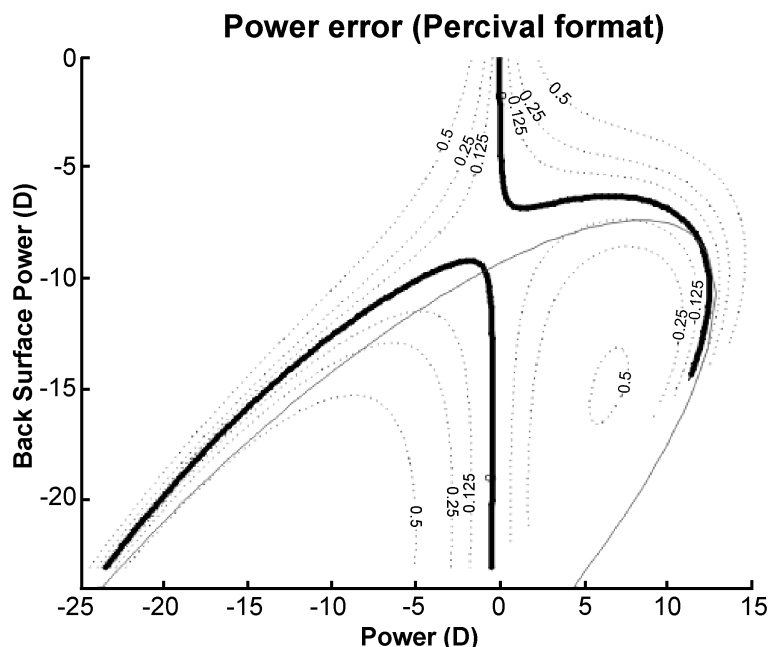


Fig. 6 Map of the mean oblique error in terms of the total power of the lens and the back surface power. The bold solid line represents the formats having null aberration. The thin solid line corresponds with the academic solution provided by the Tscherning ellipse.

The two facts related with spherical surfaces—limited range of power for which oblique errors can be corrected, and pronounced bending, even with the Ostwalt form—have led to the onset of aspheric surfaces in the ophthalmic industry.^[8]

Lenses with Aspheric Surfaces

Aspheric surfaces were introduced in the ophthalmic industry to overcome the limitations of spherical surfaces.^[38–40] The most widely used family of aspheric surfaces are the *conicoids*. They are surfaces with revolution symmetry obtained by turning around a symmetry axis of any of the classical conic curves: ellipse, parabola, and hyperbola—giving rise to the concept of ellipsoid, paraboloid, and hyperboloid, respectively. Assuming that the z -axis coincides with the symmetry axis, the equation describing a conicoid is

$$z = R \left(1 - \sqrt{1 - p \frac{x^2 + y^2}{R^2}} \right)$$

where z is the sagittal value at coordinates (x, y) , R is the curvature radius at the vertex, and p is the asphericity parameter. When $p < 0$, the surface is a hyperboloid. When $0 < p < 1$, the surface is a prolate ellipsoid. $p = 1$ stands for a sphere and $p > 1$ stands for an oblate ellipsoid.

Finally, if the limit $p \rightarrow 0$ in the previous equation is taken, we obtain

$$z = \frac{x^2 + y^2}{2R}$$

which is the equation of a paraboloid. The main geometrical characteristic of the conicoids is that tangential curvature radius of paraboloids, hyperboloids, and oblate ellipsoids (prolate ellipsoids) become larger (smaller) than the main curvature radius at the vertex, as the observer moves farther away from it in the radial direction. The increase of the tangential curvature radius produces a reduction of the tangential power so that OAE can be compensated for with flatter lens forms when the front surface is substituted by a paraboloid, hyperboloid, or oblate ellipsoid. The oblate ellipsoid can be used to compensate for astigmatic error in high-powered lenses if the designed will grind it on the back surface, where the increased back tangential power will compensate for the front tangential power of the front spherical surface (this was the design for the Katral lens by Zeiss, the first aspheric introduced for aphakia and high hypermetropia).

The first aspheric designs used in ophthalmic optics were all intended to compensate for oblique errors in high-powered positive lenses. In 1950, W. Merté introduced a kind of plano-aspheric lenses known as Merté's conicoidal lenses. For each back vertex power, asphericity parameter of the back surface can be exactly calculated in order to eliminate the OAE for a determined sight angle.



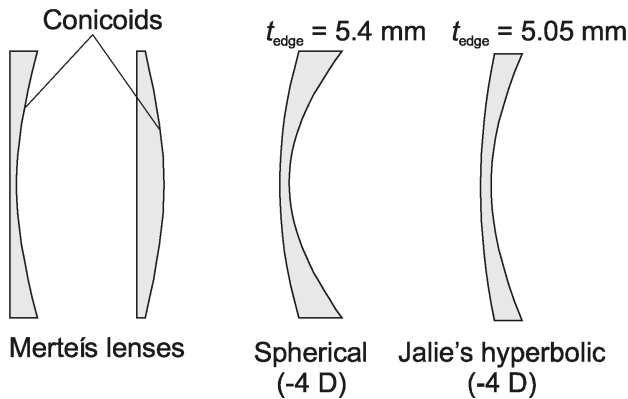


Fig. 7 Aspheric formats for ophthalmic lenses.

In 1981, M. Jalie patented the Jalie's hyperbolic lenses for which the surface of greater curvature (front surface in positive and back surface in negative lenses) is substituted by a hyperbolic surface. Jalie's design allows for flatter and thinner lenses for each available power, while keeping optimum oblique performance (Fig. 7).

Nowadays, aspheric lenses are available for medium- and even low-powered lenses, both positive and negative. New aspheric surfaces beyond conicoids have been introduced in the ophthalmic industry—for example, the polynomial surfaces or those described via *sp*-lines. Typical applications of these surface descriptions are blended lenticulars, for which there is an aspheric, smooth transition from the curvature radius of the optical zone to the outer rim of the lens.

As a general rule, aspheric designs of any power allows for flatter, thinner, and lighter lenses in any range power, while keeping oblique errors and distortion equal or smaller than those in spherical lenses.

MANUFACTURING OF OPHTHALMIC LENSES

In the last century, the ophthalmic industry has been influenced by a number of characteristics that did not share common elements with other branches of optics. A lens manufacturer had to attend to two important requirements: an enormous variety of lenses and a very short delivery time.^[41,42]

Lenses may be made from different materials, about ten; they can be colored with a complete range of tints, for each material and tint, they should be available in different geometries (aspheric and spherical) and different diameters. Also, aside from all the previous variables, there will be bifocal (two to six different designs, depending on the manufacturer), trifocal, and progressive lenses with addition values from 0.25 to 4 D in 0.25-D

steps. Finally, for each set of material, tint, geometry, and type (monofocal or multifocal), there will be a full range span of powers, typically from -25 to $+20$ D in 0.25-D steps, and for each spherical power, different values of astigmatism, typically from 0 to 4 D in 0.25-D steps. When a practitioner asks the manufacturer for a determined lens prescription, this should be delivered in a few hours. In some prescription laboratories, lenses within a smaller range of powers are delivered in just one hour, making it possible for the patients to have their spectacles finished in just one session. All this variability and capacity of quick fulfillment is only achieved via a proper combination of stock and a very well-designed and established work flow. Lens design is a key parameter to combine product quality, variability, and short delivery time.

The first family of corrected lenses (point-focal) was the Punktal series designed by M. von Rhor in 1908, and patented by Zeiss in 1911. Each lens was calibrated to its optimum form in order to eliminate oblique astigmatism. Nowadays, this approach is untenable because a large quantity of tools should be necessary for manufacturing the lenses in a typical modern catalogue. Instead of this approach, corrected curve lens series are employed. Then, the practical realization of the design of ophthalmic lenses uses the tolerance of the eye to a small amount of aberration very efficiently. As can be seen in Fig. 6, the formats having zero aberration correspond with unique combinations of total power and back (or front) surface power. However, if a small amount of aberration is allowed, the formats are located in an area that could be used to reduce the costs of manufacturing ophthalmic lenses by creating series or families of corrected curves.

In these series, the full range power is divided into intervals. All the lenses within an interval share the same base curve—that is, they have the same (usually front) surface. With this approach, optimum lens form is not achieved, but the power intervals can be selected so that the maximum aberration within each interval is below a predefined tolerance. The larger the number of intervals of the corrected curve lens series, the smaller the tolerance will be. This concept of corrected curve is understood by analyzing the plot of the aberration. For example, in Fig. 8, an area with a given amount of tolerable aberration needs to be fixed. For a given refracting error, all the formats lying in a vertical segment inside this area will compensate for the ametropia showing a controlled amount of aberration. In an equivalent way, one can include those formats sharing the same surface power of the back surface as a horizontal segment. As can be observed, these formats include a range of refracting powers, or refracting errors. The total range of powers of the ophthalmic lens can be paved by adding horizontal segments inside the area of constrained aberration. These segments represent the corrected curves of manufacturing series. They allow the

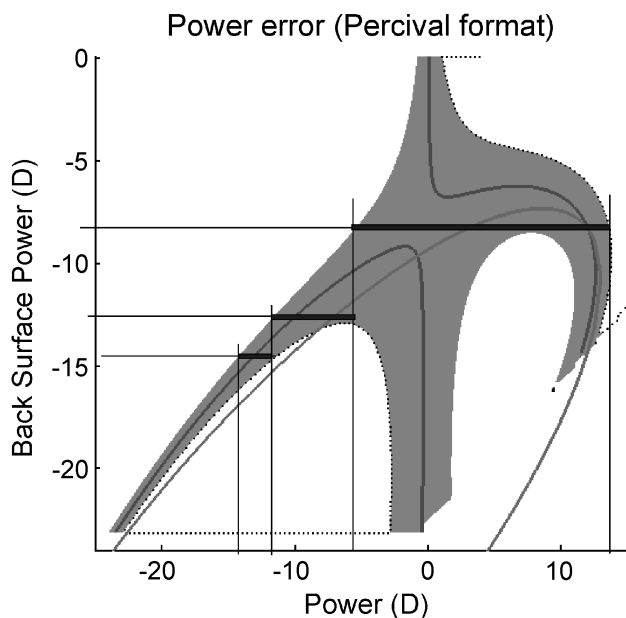


Fig. 8 Construction of the semifinished blanks for a given family of ophthalmic lenses having a controlled amount of mean oblique error (power error).

designer to finish one of the surfaces with a given surface power and apply it to a prescribed range of total powers. It is interesting to note that the corrected curves obtained for controlling an aberration in near or far vision can or can not control the other aberration in the other regime. Thus, depending on the design parameters, it is possible to tailor a specialized family of lenses.

This lens design approach allows the manufacturing process to be divided into two steps: first, in a mass production factory, semifinished lenses are ground with the predefined base surfaces. These semifinished lens blanks are sent to a prescription laboratory where they are stocked. When a practitioner places an order to the prescription laboratory, they pick up the semifinished lens with the appropriate base curve and then grind the other surface (usually the back one). There are also a small percentage of lenses finished at the mass production factory, and stored at the prescription laboratories. Either these lenses are very cheap to manufacture (for example, molded polycarbonate lenses), or they are easily stored (for example sunglasses).

Merit Functions

Ophthalmic lenses for compensating spherical ametropies, although simple optical systems, are very well adapted to the performance of the eye. However, in the dispensing and fitting of these lenses, there are some other aspects that need to be included in the design aside, from

optical performance. These aspects can be aesthetic (curvature of the front surface, presence of power rings in negative lenses, etc.), ergonomic (thickness, weight, distance of wearing), manufacturing-related (maximum diameter, thickness of the rim, etc.), related with the intended use of the lens (for far vision, or near vision), or directed to compensate some other anomalies of the vision that an optometrist may indicate. All these design elements, along with the optical performance of the lens, are included in a merit function that is minimized within the design process. These merit functions share a similar structure with those employed for precision optics, but are properly adapted for ophthalmic use. For example, the difference between the actual oblique power error and the target do not enter squared into the merit function, as the practice goes with other aberrations. This is because the sign of power error is important. Positive power error can be compensated by a small accommodation by hyperopic persons, while myopic persons can compensate for negative power errors. Finally, the output of the merit function optimization must be discretized into corrected curve lens series, as previously explained.

This is the actual process practiced in the ophthalmic lens design nowadays. Different designs are presented to the user, in terms of their relative performance, to comply with the expectations of the wearer of the lens. This information should be available to the optometrist and the dispensing professional to assist the patient and the user in choosing the optimum ophthalmic lens.

LENSES FOR PRESBIOPIA

There are many different ways to compensate presbyopic patients. These options include near-vision monofocal lenses (which must be alternated with distance vision lenses when the presbyopic also happens to be ametropic), bifocal lenses, trifocal lenses, progressive addition lenses, and monofocal lenses with increased depth of focus, which only became available in the last five years. Each type of lens is calibrated to suit specific presbyopia conditions, age, and patient activities. Monofocal lenses have the advantage of a wide field of view (limited by the frame) and good optical quality, as the lens belongs to an excellent lens series. However, ametropic persons will have to alternate between two spectacles when passing from distance vision to near vision. Patients with a very short accommodation range will even need a further spectacle for intermediate vision.

Bifocals and Trifocals

Bifocals have two regions with different powers; one region, usually at the upper portion of the lens, for distance



vision, and another region at the bottom for near vision, which is called the segment. Usually, the near region is made by a local modification of curvature or refraction index in the lens for distance vision, which is called the main lens. Since the invention of bifocals in 1785 by B. Franklin, there have been many different designs, usually depending on the availability of adequate manufacturing technology. Nowadays, there are two main technologies:

One-piece solid bifocals. The near region is generated by a change of curvature of the front surface. Solid bifocals may be further subdivided into invisible (without steps in the line which separate the two regions), blended (with smooth transitions between regions), and executive styles (with straight edge).

Fused bifocals. The near region is generated by a piece of glass with higher refraction index inserted in the main lens.

Bifocals allow the presbyopic patient immediate change from distance to near vision, but they have some limitations.

Prismatic effect is difficult to control in the near region. When the shape and size of the near zone is fixed, very little degree of freedom is left to properly locate the optical center of the near portion.

Some types of bifocals (mainly those with round segment) present image jump, which is a sudden change in the prismatic effect caused by the addition lens. Image jump produces scotoma, and makes adaptation to bifocal lenses difficult.

The separation line between distance and near vision may produce an unsightly lens. Also, stepped separation lines may produce light dispersion and/or parasite reflections that may lead to user annoyance.

Aberrations in the near portion become larger than those in monofocal lenses. Usually, the lens form is selected for optimizing distance vision. In solid bifocals the curvature of the front surface in the near portion grows, so best form condition is lost. In fused bifocal, a different material and a third surface are introduced in the near portion, so best form condition is also lost.

Prismatic effect and image jump can be easily estimated within the paraxial approximation. From a theoretical point of view, the bifocal may be considered as the superposition of two thin lenses: the main lens for distance vision and optical center at L , and a smaller lens with positive spherical power (without cylinder) and optical center at O , which contribute with the addition. It is convenient to express the optical characteristics of the near portion in terms of the addition, A , and the location of its optical center, O , with respect to the optical center of the main lens. Matrix Prentice's law is used to obtain the following formula for the computation of prismatic

effects at any point Q located in the near portion of a bifocal lens^[43]

$$p_Q = -(\mathbf{F} + \mathbf{A})\mathbf{r}_{LQ} + A\mathbf{r}_{LO}$$

where p_Q is the prismatic effect at Q , $\mathbf{F}$ is the dioptric power matrix of the distance portion, $\mathbf{A}$ is the matrix of the addition, and $\mathbf{r}_{XY}$ is the position vector of point Y relative to point X . The first term in the previous formula correspond with the prismatic effect of a lens with the total near power, and the second term depends on the addition and the type of bifocal through the distance $\mathbf{r}_{LO}$. The optical center of the near region C , is easily found by making the prismatic effect to be zero. The result is

$$\mathbf{r}_{LC} = (\mathbf{F} + \mathbf{A})^{-1}A\mathbf{r}_{LO}$$

The image jump, Δp , at any point K located on the separation line is simply given by

$$\Delta p = A\mathbf{r}_{OK}$$

In general, there is a trade-off between image jump and the amount of prismatic effect in the near portion, both being determined by the position of the optical center of the addition. Round segments produce prismatic effects in the near region similar to those in monofocals, whereas they present the largest values of image jump. In contrast, segments with straight edge present the larger prismatic effects and smaller or negligible image jumps.

When the amplitude of accommodation decreases to less than 1 D, clear vision at intermediate distances is compromised with bifocals. To help the patient manage this situation, trifocal lenses were invented by J.I. Hawkins in 1826.^[6] They have two segments: one for intermediate vision with half the addition, and a second one for near vision with full addition. Trifocals are made with the technology for fused bifocals. Previous equations are easily generalized to compute image jumps and prismatic effect in trifocal lenses. The main drawbacks of trifocal lenses are decreased fields of view, annoying separation lines, and unsightly aspect.

Progressive Power Lenses

Progressive power lenses were designed and introduced in an attempt to improve the performance of previous multifocal (trifocal and bifocal lenses) variants. The objectives were clear vision at all distances, elimination of separation lines, and a more aesthetic lens.^[6,8,44] The first progressive lens with commercial success was the Varilux 1, which was designed by B. Maitenaz, with Essilor, in 1958. Increase of power from the upper portion to the bottom of the lens is achieved by progressively increasing the curvature radius of the front surface. In Varilux 1, the half upper part of the surface was



spherical. Some millimeters below the point for distance vision, the curvature of the horizontal sections begin to increase smoothly. The region in which the curvature increases is called the progression zone. At the bottom of this region, power stabilizes in an area named reading or near portion. It can be demonstrated that progressive variation of the surface curvature must introduce surface cylinder, which will turn into astigmatic aberration. To avoid it, Varilux 1 was designed so that all the points along the vertical central meridian were umbilical (locally spherical). As the visual point separates along the horizontal direction from the umbilical line in the progression region, astigmatism appears. Minkwitz showed that for Varilux 1 progressive lenses, the increase of cylinder, ΔC , when the visual point moves a distance Δx from the umbilical line, satisfies the relation

$$\frac{\Delta C}{\Delta x} \approx 2 \frac{\Delta S}{\Delta y}$$

where ΔS is the increase of spherical powers as the visual point moves a distance Δy along the umbilical line. According to this statement, the larger the rate of power increases, the quicker the astigmatism will grow along the horizontal direction in the progressive region. In the distance and near region, where power is stabilized, $\Delta S/\Delta y = 0$, and no astigmatism occurs. Generally speaking, for a given addition, a short progression will lead to high values of lateral astigmatic aberration but they are constrained to a small lens region. A long progression length will produce smaller and smoother increases of lateral aberration, but are spread in a wider lens area.

Since the launch of Varilux 1, there has been five generations of progressive lenses from Essilor. When patents of Varilux 2 (launched in 1972) expired, many other manufacturers have entered the market with competitive designs. In Varilux 2, aspherization of surface horizontal meridians was introduced. This gave designers more degrees of freedom to reduce lateral astigmatism beyond the imposition of Minkwitz rule. Lens asymmetry was also introduced to take into account the convergence of the visual points when going from distance to near vision. This was implemented by introducing a continuous displacement (inset) of the umbilical line to the nasal side along the progression zone. Newer designs came out successively to address various problems.^[45] variation of the inset as a function of age and/or ametropia, improvements of the binocular behavior of the pair of progressive lenses, reduction of lateral astigmatism, reduction of gradients of sphere and astigmatism, improvement of the lens ergonomics, and quite recently, reduction of the swimming effects as a consequence of differential distortion of the extrafoveal field of view. The search for

better progressive lenses has stimulated active research on ophthalmic optics and vision in the last 15 years. Many works have been published concerning the measurement of these lenses, as well as the physiological consequences derived from their use.^[45–50]

CONCLUSION

Ophthalmic lenses are the most widespread optical elements in the world. However, simplicity in their design needs to be adapted to the anatomical and physiological constraints imposed by the eye. The design needs to improve optical performance in terms of minimization of optical aberrations involved—mainly field curvature and oblique astigmatism. At the same time, the results should also consider aesthetic and ergonomic criteria. Practical manufacturing of ophthalmic lenses also utilizes tolerance of the eye to provide an efficient way of grinding, polishing, and finishing lenses in order to satisfy strong demands on variety and short delivery times. New designs using aspheric surface help to improve the performance of monofocal lenses. At the same time, progressive lenses for presbyopia represent a step forward in the complexity and performance of optical elements having only two refracting surfaces. Knowledge of the compound eye–lens system, along with the basis of the design of ophthalmic lenses, must be used in the final adaptation of the ophthalmic lens for a given user.

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